On the Theory of Abelian Groups

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Among *finite groups*, those whose operations are *interchangeable*, the so-called *commutative* or *Abelian groups*, are distinguished by particular simplicity and transparency. This transparency rests on the fact that we are able to characterize the different commutative groups by definite numbers standing in very simple relation to them, their "*invariants*." From every *Abelian group* one can, namely, select a *system of generating operations*

$$A_1, A_2, \ldots, A_n$$

which has the following properties: 1. Every operation can be represented, and in only one way, as a product of powers of the operations $A_1, \ldots, A_n$. 2. Of the orders $a_1, a_2, \ldots, a_n$ of the operations $A_1, A_2, \ldots, A_n$, each is divisible by the following one. The numbers a are thereby completely determined by the group, and in turn determine the group (if holohedrically isomorphic groups are not regarded as different). They are called the "*invariants*" of the group. Naturally the system of invariants of the group may be prolonged arbitrarily by adding invariants of value 1, since one may add the identity as often as desired to the system of generating operations. The number of invariants different from 1 is called the "rank" of the group and indicates how many operations are required to generate it.

In order that in a commutative group $\alpha = (a_1, a_2, \ldots)$, that is, a group α with invariants $a_1, a_2, \ldots$, there be present a subgroup isomorphic with

$$\beta = (b_1, b_2, \ldots),$$

it is necessary and sufficient that the quotients

$$\frac{a_1}{b_1}, \frac{a_2}{b_2}, \ldots$$

be integers.

With these theorems, and with investigations which are readily dealt with by their aid, the theory of *Abelian groups* is by no means exhausted; further occupation with them soon leads to questions which require entirely new investigations. One such question concerns the *composition of groups from given factors* in the sense first given by *C. Jordan.* Let

$$\gamma = (c_1, c_2, \ldots), \qquad \alpha = (a_1, a_2, \ldots), \qquad \beta = (b_1, b_2, \ldots)$$

be *Abelian groups.* We shall say: "γ is contained in the *product* $\alpha \cdot \beta$" if γ possesses a subgroup α' isomorphic with α, for which moreover γ/α' is isomorphic with β. If, then, by the *product* $\alpha \cdot \beta$ one understands the totality of the groups contained in $\alpha \cdot \beta$, the investigation easily shows that, for this formation of products, the *commutative* and *associative* laws do hold, but that by no means does an equation of the form $\alpha \cdot \beta = \alpha \cdot \beta'$ imply equality of the groups β and β'. It is nevertheless easy to arrange the calculation with commutative groups so that the formal calculating operations have validity throughout. One arrives at this by sharpening the question whether a group γ is contained in the product $\alpha \cdot \beta$ to the following: *How often is the group γ contained in $\alpha \cdot \beta$*, that is, how many subgroups α' does γ possess of such a kind that α' is isomorphic with α and γ/α' with β? If this number is denoted by $t(\gamma; \alpha, \beta)$, it is natural to define a symbolic calculation with groups by the equation

$$\alpha \cdot \beta = \sum_{\pi} t(\pi; \alpha, \beta) \cdot \pi,$$

which represents the *product* of two groups as a *sum* of groups, whose summands are the groups π contained in the product $\alpha \cdot \beta$ in the sense just indicated, and each occurs as many times as it is contained in the product. On account of the easily proved relation

$$t(\gamma; \alpha, \beta) = t(\gamma; \beta, \alpha),$$

the commutative law holds for this calculation. It follows further from an equation of the form $\alpha \cdot \beta = \alpha \cdot \beta'$ that $\beta = \beta'$. Every *integral function of groups* can, on the basis of the defining equation, be represented as a *linear homogeneous function* of groups, and in *only one* way.

But despite these elegant properties of the commutative groups, one is still very far from answering the cardinal question concerning the groups contained in a product $\alpha \cdot \beta$. This calls for deciding, by a general theorem, the vanishing or non-vanishing of the function $t(\gamma; \alpha, \beta)$. (In each individual given case, of course, the decision can be supplied by a finite procedure.) For various special cases, however, many remarkable results arise. If, for example, of the groups α, β one, say β, is of rank 1, so that only its first invariant b_1 has a value different from 1, then the groups contained in $\alpha \cdot \beta$ can be specified easily. Let

$$\alpha = (a_1, a_2, \ldots), \qquad \gamma = (c_1, c_2, \ldots).$$

Then the following are necessary and sufficient conditions that γ be contained in $\alpha \cdot \beta$:

1. $c_1 \cdot c_2 \cdot c_3 \cdots = a_1 \cdot a_2 \cdot a_3 \cdots b_1$,
2. The quotients $\frac{c_1}{a_1}, \frac{a_1}{c_2}, \frac{c_2}{a_2}, \frac{a_2}{c_3}, \frac{c_3}{a_3}, \frac{a_3}{c_4}, \ldots$ must be integers.

This theorem can also be expressed in another form: in order that a group $\alpha = (a_1, a_2, \ldots)$ may be extended by adjoining a single generating operation to a group $\gamma = (c_1, c_2, \ldots)$, it is necessary and sufficient that the quotients

$$\frac{c_1}{a_1}, \frac{a_1}{c_2}, \frac{c_2}{a_2}, \frac{a_2}{c_3}, \ldots$$

be integers. – Also for the case in which, of the groups α, β, one is only of rank 2 (the other arbitrary), I have determined the groups contained in $\alpha \cdot \beta$.

A further special case: of the groups

$$\alpha = (a_1, a_2, a_3), \qquad \beta = (b_1, b_2, b_3), \qquad \gamma = (c_1, c_2, c_3),$$

none is of higher than third rank. In order that γ be contained in $\alpha \cdot \beta$, it is necessary and sufficient that:

1. $c_1 \cdot c_2 \cdot c_3 = a_1 \cdot a_2 \cdot a_3 \cdot b_1 \cdot b_2 \cdot b_3$,
2. the quotients

$$\begin{array}{llll} \frac{c_1}{a_1 \cdot b_1}, & \frac{a_1 \cdot b_3}{c_1}, & \frac{a_3 \cdot b_1}{c_1}, & \\ \frac{c_2}{a_1 \cdot b_2}, & \frac{c_2}{a_2 \cdot b_1}, & \frac{a_2 \cdot b_3}{c_2}, & \frac{a_3 \cdot b_2}{c_2}, \\ \frac{c_3}{a_1 \cdot b_3}, & \frac{c_3}{a_2 \cdot b_2}, & \frac{c_3}{a_3 \cdot b_1}, & \frac{a_3 \cdot b_3}{c_3} \end{array}$$

must be integers.

In these investigations one may, as is easy to see, restrict oneself to considering those *Abelian groups* whose orders are powers of one and the same prime number p. Let therefore now

$$\alpha = (p^{a_1}, p^{a_2}, \ldots), \qquad \beta = (p^{b_1}, p^{b_2}, \ldots), \qquad \gamma = (p^{c_1}, p^{c_2}, \ldots).$$

It is then natural to approach the question of the vanishing or non-vanishing of the function $t(\gamma; \alpha, \beta)$ by aiming at the general determination of the function t. I have carried out such a determination with the help of recursion formulas. $t(\gamma; \alpha, \beta)$ is a function of the prime number p

and of the exponents a_i, b_i, c_i; if the exponents are regarded as definite given numbers and t as a function of p alone, the investigation teaches that t is an integral rational function with integral coefficients, whose degree certainly does not exceed the number

$$m = (c_1 - a_1 - b_1) + 2(c_2 - a_2 - b_2) + 3(c_3 - a_3 - b_3) + \cdots .$$

If, therefore, the coefficient of p^m is denoted by $h(\gamma; \alpha, \beta)$, then $h(\gamma; \alpha, \beta)$ is independent of p and cannot be negative, because otherwise for large values of the prime number p the function $t(\gamma; \alpha, \beta)$ would take negative values, contrary to its nature as a number. – The form, however, in which the function $t(\gamma; \alpha, \beta)$ presents itself is not suited to decide in general when it vanishes identically (that is, with the exponents a_i, b_i, c_i fixed, for every value of p), and when it does not. According to the investigations up to now, the following theorems may be regarded as very probable: 1. *If the function $t(\gamma; \alpha, \beta)$ does not vanish identically, then it vanishes for no value of the prime number p.* 2. *The function $t(\gamma; \alpha, \beta)$ either actually rises to degree m, or else it vanishes identically altogether.* Accordingly, the vanishing and non-vanishing of the function $t(\gamma; \alpha, \beta)$ would be quite independent of p and conditioned solely by the vanishing or non-vanishing of $h(\gamma; \alpha, \beta)$.

The coefficient $h(\gamma; \alpha, \beta)$ now admits a remarkable representation, on which a brief word may be said here. For this purpose let us think of a natural number n chosen arbitrarily large but fixed, and then consider exclusively such groups

$$\alpha = (p^{a_1}, p^{a_2}, \ldots, p^{a_n})$$

whose rank is $\leq n$, and assign to each such group a definite *symmetric function of n indeterminates* $x_1, x_2, \ldots, x_n$ as follows. The exponents $a_1, a_2, \ldots, a_n$ satisfy the inequalities $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$; if one therefore puts

$$a'_i = a_i + n - i,$$

then among the a' there hold the inequalities

$$a'_1 > a'_2 > \cdots > a'_n \geq 0.$$

The expression

$$S'(\alpha) = \sum \pm x_1^{a'_1} x_2^{a'_2} x_3^{a'_3} \cdots x_n^{a'_n},$$

whose terms arise from $x_1^{a'_1} x_2^{a'_2} \cdots x_n^{a'_n}$ by subjecting the exponents to all permutations, with the sign $+$ or $-$ to be taken according as the permutation belongs to the even or odd class, represents an *alternating function* of $x_1, \ldots, x_n$. We shall call the alternating functions $S'(\alpha)$ corresponding to the groups α in this way *simple alternating functions.* It is then easy to see that every alternating function can be expressed, and in only one way, linearly and homogeneously in terms of simple alternating functions. Further, let D denote the difference-product of $x_1, \ldots, x_n$, and let the symmetric function

$$S(\alpha) = \frac{S'(\alpha)}{D}$$

be called a *simple symmetric function.* It is then clear that every symmetric function can be expressed, and in only one way, linearly in terms of simple symmetric functions. If now one expresses the product $S(\alpha) \cdot S(\beta)$ of two simple symmetric functions linearly in terms of simple symmetric functions, one obtains the remarkable result

$$S(\alpha) \cdot S(\beta) = \sum h(\gamma; \alpha, \beta) \cdot S(\gamma),$$

where the sum extends over those groups γ whose rank is $\leq n$, and the coefficients $h(\gamma; \alpha, \beta)$ have the meaning given above. Accordingly everything would come down to deciding which coefficients in the expansion of $S(\alpha) \cdot S(\beta)$ turn out to be > 0. Whether the answer to our question is made easier thereby must, of course, remain undecided. Up to now, my own proof that negative coefficients can never occur in the expansion of $S(\alpha) \cdot S(\beta)$ has not been possible otherwise than by the detour through the theory of *Abelian groups.*